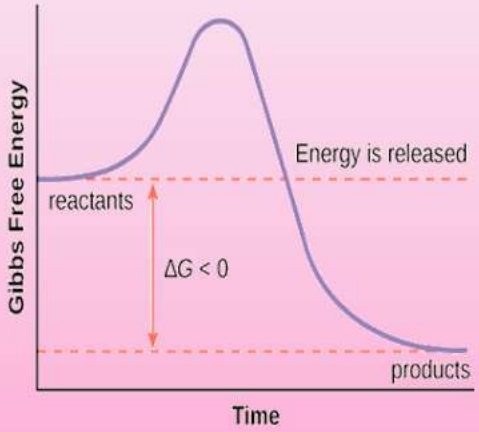
CHEM 1043 General Chemistry II

Topic 5

Spontaneous Change



Dr. A T L LI

What to learn in this topic?



Spontaneous Change

- ✓ Qualitative cation analysis Entropy Change: Clausius's View and Boltzmann's View
- ✓ Combining Clausius's and Boltzmann's Ideas: Absolute Entropies
- ✓ The Second Law of Thermodynamics
- ✓ Gibbs Energy Change of a System of Variable Composition: $\Delta_r G°$ and $\Delta_r G$
- ✓ $\Delta_r G°$ and K as Functions of Temperature
- ✓ Using $\Delta_r G$ to Describe a Reaction
- ✓ Coupled Reactions

Your prior knowledge related to this topic

CHEM1042 General Chemistry I

Topic 3 Thermodynamics

- ✓ Heat and Work
- ✓ The First Law of Thermodynamics
- ✓ Internal Energy and Enthalpy
- ✓ Types of Thermodynamics Processes
- ✓ Standard Enthalpy Changes
- ✓ Hess's Law
- ✓ Born-Haber Cycle
- ✓ Bond Dissociation Enthalpy
- ✓ Introduction to Entropy
- ✓ The Second and Third Law of Thermodynamics
- ✓ Introduction to Gibbs Free Energy

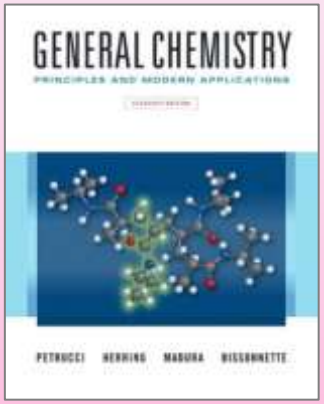


Your reading:

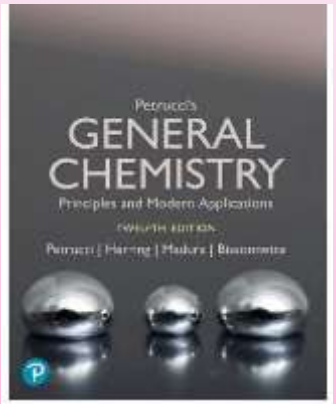
Petrucci; Herring; Madura; Bissonnette:
General Chemistry: Principles and Modern Applications, 11th or 12th edition, Pearson

Chapter 7) Thermochemistry

Chapter 13) Spontaneous Change: Entropy and Gibbs Energy



11th edition



12th edition

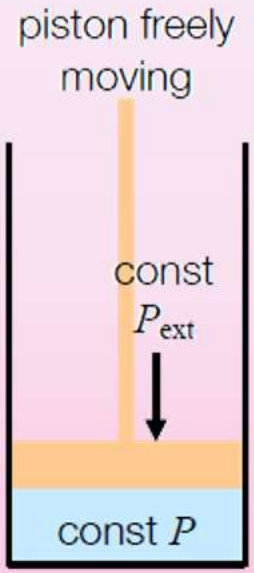
Part 1: Thermodynamics (Extension)



Types of Thermodynamic Processes (Revision)

1. **Isobaric process** (constant P)

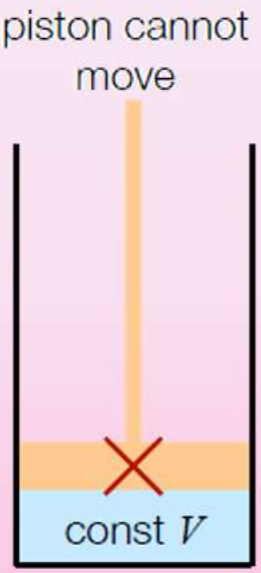
- $w = -P_{ext}\,\Delta V = -nR\Delta T$
- $q_p = \Delta H_p = C_p\Delta T$
- $\Delta U_p = q_p + w = C_p\Delta T - P_{ext}\,\Delta V$



isobaric process

2. **Isovolumetric process** (constant V)

- $w = 0$ (no work done is possible)
- $q_v = \Delta U_v = C_v\Delta T$
- $\Delta H_v = \Delta U_v + \Delta(PV) = C_v\Delta T + V\Delta P$
  $= \Delta U_v + nR\Delta T$



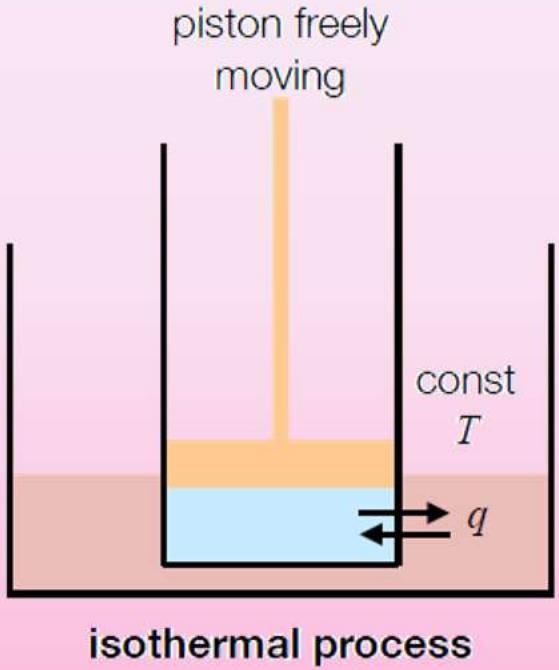
isovolumetric process

Types of Thermodynamic Processes (Revision)

3. **Isothermal process** (constant T)

- $\Delta U = 0$ (for ideal gases)
- $\Delta H = \Delta U + \Delta(PV) = 0$
- $q = -w$

(work done in the expansion exactly compensated for by the heat flowing in)

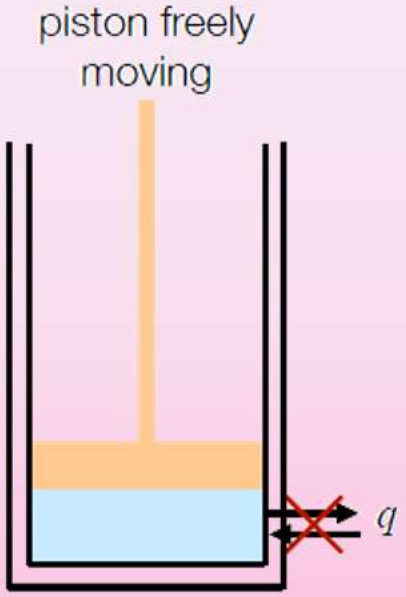


isothermal process

4. **Adiabatic process** (no heat transfer)

- $q = 0$ (no heat transfer is possible)
- $\Delta U = w = C_v \Delta T$
- $\Delta H = \Delta U + \Delta(PV) = V\Delta P = C_p \Delta T$

(increase in internal energy comes from work done by surrounding)



adiabatic process



Class Practice 1

Isovolumetric Process

Consider 1.00 mol of an ideal gas with $C_V = 3/2$ R that undergoes a temperature change from 125 K to 255 K at a constant volume of 10.0 L. Calculate ΔU, q, and w for this change.



Class Practice 2

Isobaric Gas Expansion

Consider 1.00 mol of an ideal gas with C_p = 5/2 R that changes temperature change from 125 K to 255 K at a constant pressure of 10.0 atm. Calculate ΔU, ΔH, q, and w for this change.

Work for Reversible Pathways

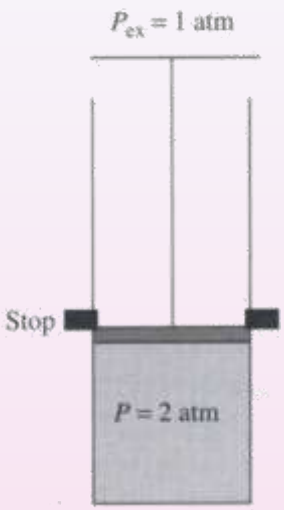
- As briefly mentioned in CHEM1042, there are two pathways, reversible and irreversible paths, to which thermodynamics can be applied.
- The thermodynamics problems you have learnt so far all follow irreversible pathways. To calculate the work done for an irreversible path, you use $w_{irrev} = -P_{ext}\Delta V$.
- In the following, you will see how we calculate work done for reversible pathways as well.

Reversible vs Irreversible Processes

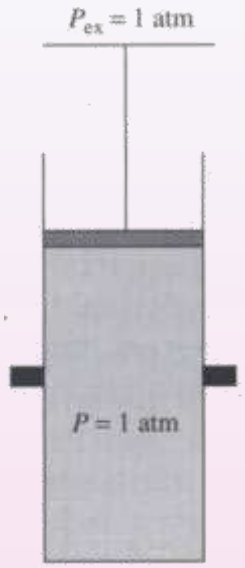
- If an infinitesimal change in the driving variable does not change the direction of the process, the process is **irreversible.**
- When an **infinitesimal** opposing change in the variable that drives the process in reversal direction, the process is **reversible**.

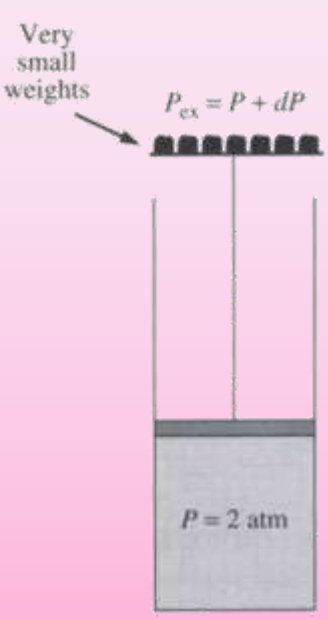
Reversible & Irreversible Processes

Example: Volume changes



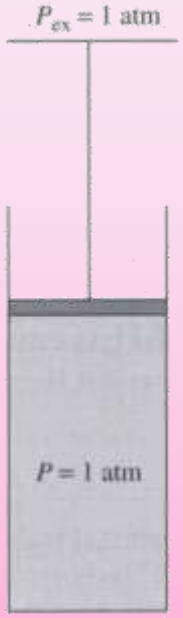
Irreversible expansion by removal of stops





an infinitesimal change each time

Reversible expansion by removal of weights, one at a time



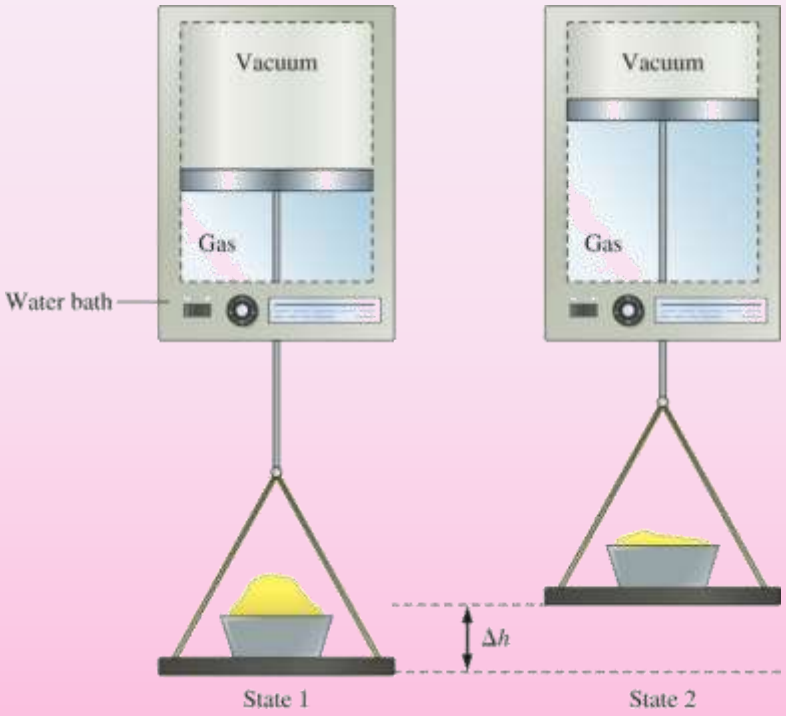
Sufficiently slow changes are a good approximation

Comparing Work for Different Paths

- Now consider the expansion from ***state 1*** to ***state 2*** (see **Fig. 1** below). The weight is equivalent amount of sand.
- Imagine sand is removed very slowly from the pile in ***state 1***, say, one grain at a time. When exactly half the sand has been removed, the gas will have reached state 2.

Figure 1 A different method of achieving the expansion of a gas

In this expansion process, the weight is a pan containing sand, which has a mass of $2M$ equivalent to that of the weights in the initial state (***state 1***). In the final state (***state 2***) the mass has been reduced to M.



Comparing Work for Different Paths

- The very slow expansion proceeds in an approximately reversible fashion. From a thermodynamics perspective, a process is ***reversible*** if it involves an infinite number of intermediate states, with the system variables changing from their initial values to their final values by **infinitesimally small** amounts.

- The total work when the system expands from V_1 to V_2 **in a reversible pathway** is the sum (integral) of all the infinitesimal changes between the limits V_1 and V_2.

$$w_{rev} = -\int P_{ex}\,dV = -\int P_{\text{int}}\,dV = -nRT\int \frac{dV}{V} = -nRT\ln\frac{V_2}{V_1}$$

- The **maximum work** that can be extracted from the same expansion process is that obtained under **reversible** conditions.



Class Practice 3

Reversible Gas Expansion

In the gas expansion process, 0.1 mol of helium gas expands from 1.02 L to 2.04 L isothermally in a **reversible** way, calculate the amount of work done for this reversible expansion.

Comparing Work for Reversible and Irreversible Processes

(I) Consider the following **irreversible** process:

❖ An ideal gas is confined in a cylinder with a weightless movable piston and the walls of the system are diathermal (→ **isothermal process**)



Original state: T, P_1, V_1 —**expansion**→ **Final state:** T, P_2, V_2 —**compression**→ **Original state:** T, P_1, V_1

(1) The gas (T, P_1, V_1) is first expanded at constant T by decreasing P_{ext} abruptly to the value P_2, where $P_2 < P_1$. *Heat flows into the system through the walls to keep the temperature constant (T). Now the state of the gas is defined by* (T, P_2, V_2), where $V_2 > V_1$.

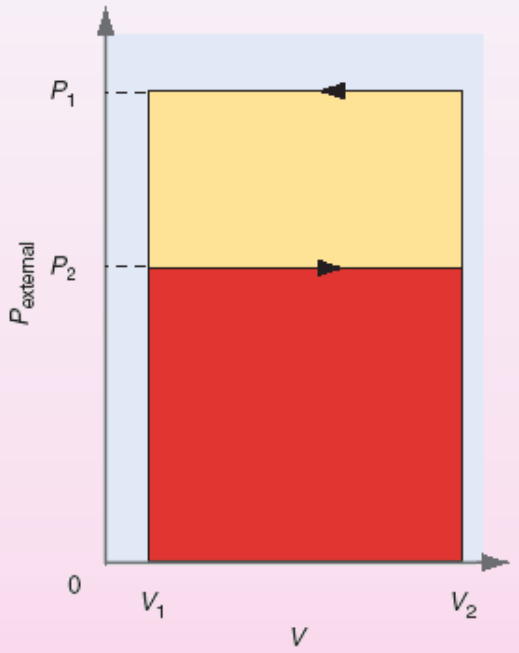
(2) The system is then returned to its original state (T, P_1, V_1) by increasing P_{ext} abruptly to its original value P_1. *Heat flows out of the system.*

Comparing Work for Reversible and Irreversible Processes

- The total work associated with this **cyclic process ($\Delta U = 0$)** is given by the sum of the work for each individual step:

$$w_{total} = \sum_i -P_{ex,i}\Delta V_i = w_{expansion} + w_{compression}$$
$$= -P_2(V_2 - V_1) - P_1(V_1 - V_2)$$
$$= -(P_2 - P_1)\times(V_2 - V_1)$$
$$> 0 \qquad \because\ P_2 < P_1 \text{ and } V_2 > V_1$$

$$\because\ \Delta U = q_{total} + w_{total} = 0 \qquad \therefore\ q_{total} < 0$$



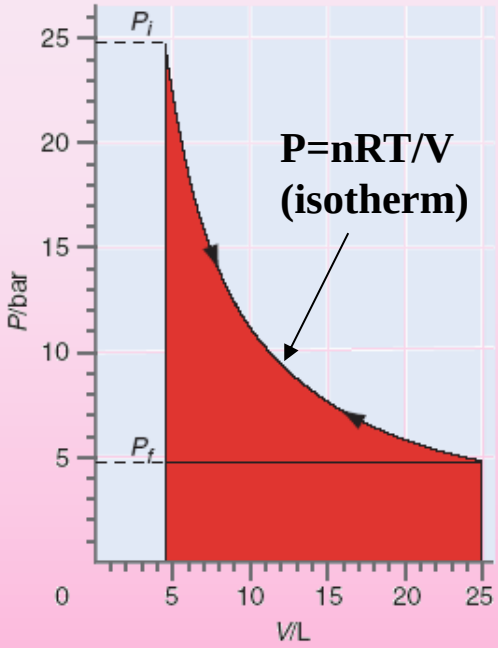
- **An indicator diagram** shows the relationship between P and V for the process and depicts the work done involved (area shows the integral $w = -\int P_{ext}\, dV$.)
- $w_{expansion}$ is given the **red** area while the $w_{compression}$ is the **total area in red and yellow**. The arrows indicate the direction of change in V in the two steps.

Comparing Work for Reversible and Irreversible Processes

(II) Consider the same process (constant T, change in P & V) is carried in a **reversible** cycle:

❖ A necessary condition: $P = P_{ext}$ at every step of the cycle.

→ P ***changes*** during the expansion and compression steps.



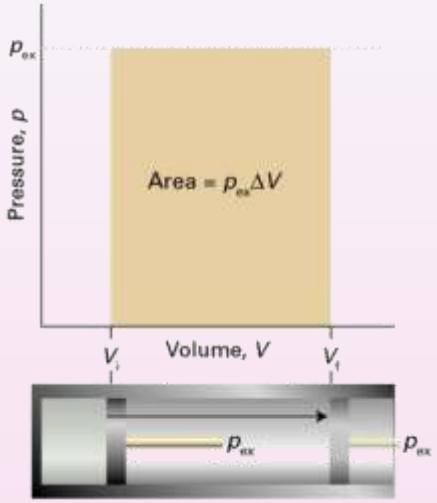
Indicator diagram for a reversible process. The work is shown as the red area.

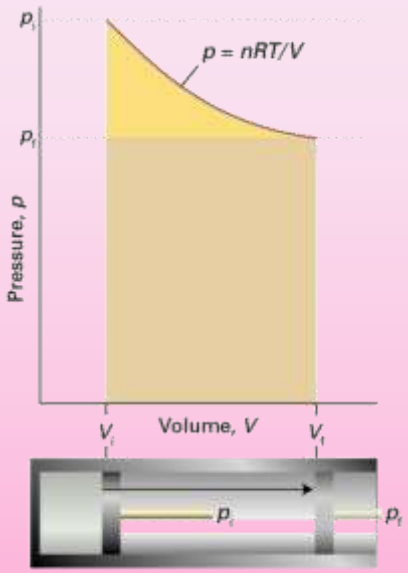
For ***expansion***, $w_{expansion} = -nRT \ln \frac{V_2}{V_1}$

For ***compression***, $w_{compression} = -nRT \ln \frac{V_1}{V_2}$

∴ The total work done for this cyclic process = 0

Comparing Work for Reversible and Irreversible Processes





Summary (Isothermal process):

- The work done by an ideal gas when it expands **isothermally against a constant P_{ext}** is equal to the shaded area.

$$w_{\text{irrev}} = -P_{ext}\Delta V$$

- The work done by an ideal gas when it expands **isothermally** and **reversibly** is equal to the area under the isotherm $P = nRT/V$.

$$w_{rev} = -nRT \ln \frac{V_f}{V_i}$$

- The work done during an irreversible expansion against the same final pressure is equal to the rectangular area.

Reversible work > Irreversible work

Comparing Work for Reversible and Irreversible Processes

Conclusion: The **maximum work** that can be extracted from the same **expansion** process is that obtained under **reversible** conditions.

Thermodynamics Properties in ***reversible isothermal process***:

$$\Delta U = q + w = 0 \qquad \Delta H = \Delta U + \Delta(PV) = 0$$

$$q = nRT \ln \frac{V_f}{V_i} \qquad w = -nRT \ln \frac{V_f}{V_i}$$

Example 1

Reversible Isothermal Gas Expansion

Calculate q, w, ΔU, and ΔH for 1.00 mol of an ideal gas expanding reversibly and isothermally at 273 K from a volume of 22.4 L and a pressure of 1.00 atm to a volume of 44.8 L and a pressure of 0.500 atm.

Suggested Solution:

In isothermal process,

$$\Delta U = q + w = 0$$
$$\Delta H = \Delta U + \Delta(PV) = 0$$

$$\begin{aligned} w &= -nRT \ln \frac{V_2}{V_1} \\ &= (1.00\,mol)\left(8.314\,\frac{J}{mol\,K}\right)(255\,K)\ln\left(\frac{44.8\,L}{22.4\,L}\right) \\ &= 1572\,J = 1.57\,kJ \end{aligned}$$

$$\begin{aligned} \Delta U &= q + w \\ &= q + 1.57 \\ &= 0 \end{aligned}$$

$$q = -1.57\,kJ$$



Class Practice 4

Reversible and Irreversible Isothermal Gas Expansion

2.00 mol of an ideal gas undergoes isothermal expansion at P_i = 25.0 bar and V_i = 4.50 L along two different paths:

(1) a single-step irreversible expansion against a constant external pressure of 4.50 bar,

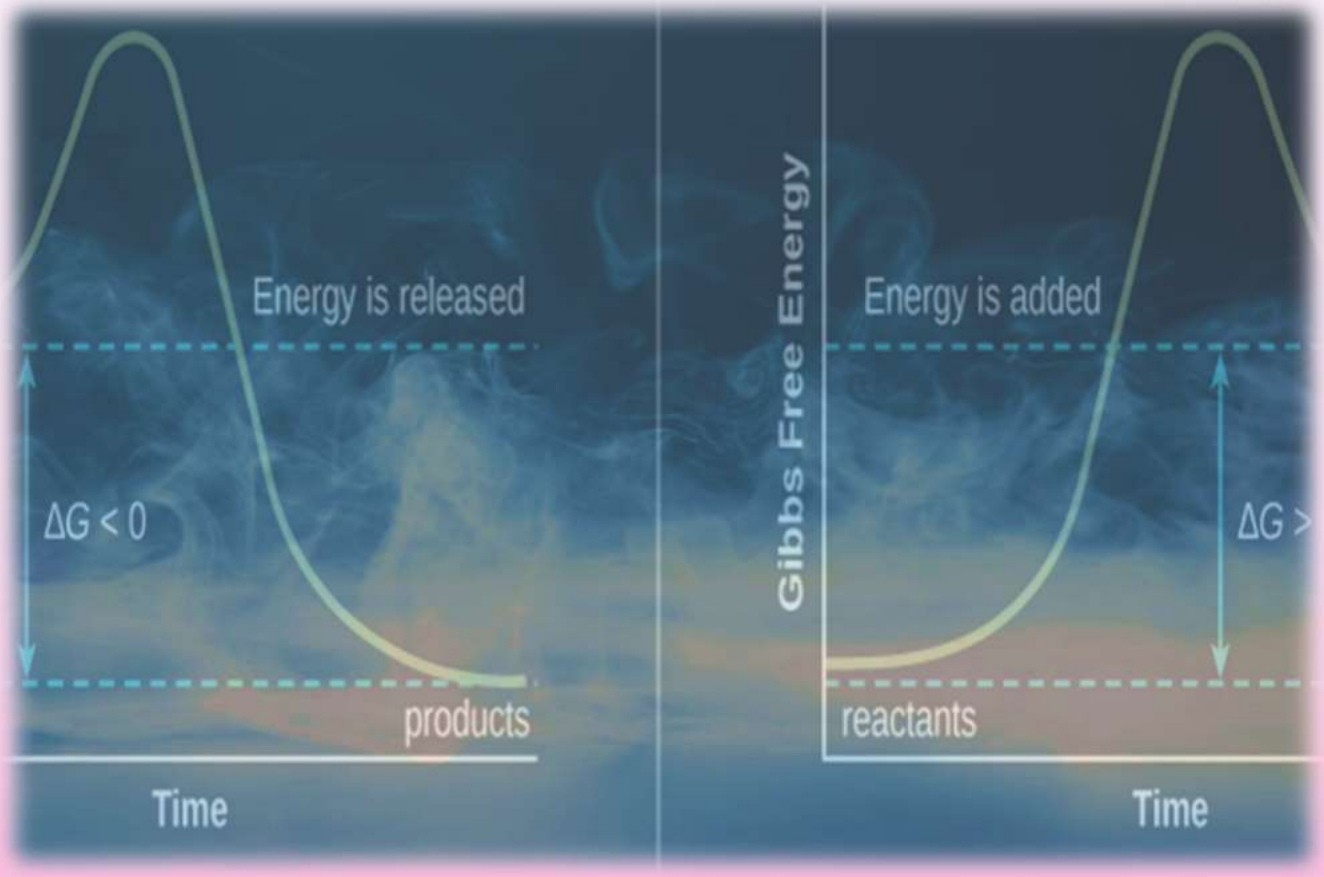
(2) reversible expansion from P_i = 25.0 bar and V_i = 4.50 L to P_f = 4.50 bar.

Calculate the work for each of these processes. For which of the irreversible processes is the magnitude of the work greater?



Class Practice 4 (Continued)

Part 2: Spontaneous Change



Spontaneity (Revision)

The Meaning of Spontaneous Change

❖ The *1st law of thermodynamics* makes no restrictions on the inter-conversion of energy in obviously unnatural ways.

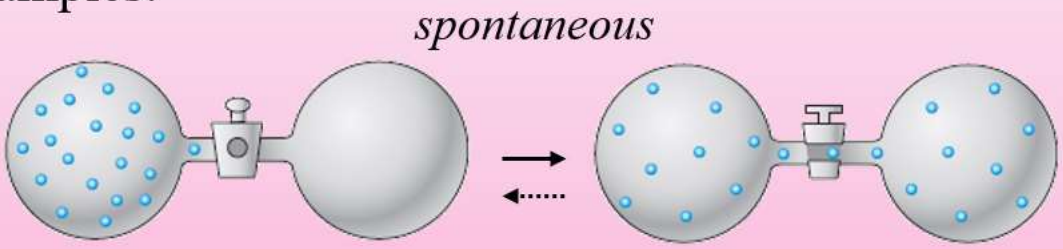
❖ All processes have a **natural direction** of change.

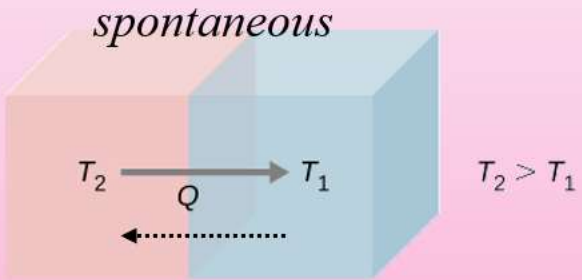
❖ A ***spontaneous*** process is one that **occurs with a natural tendency**.

❖ If a process is spontaneous, the reverse process is ***non-spontaneous*** *under the same conditions*. Non-spontaneous processes require the system to be acted on by an external agent.

❖ Once the spontaneous process is initiated, they go on their own without further intervention. Some spontaneous processes occur very slowly and others occur rather rapidly.

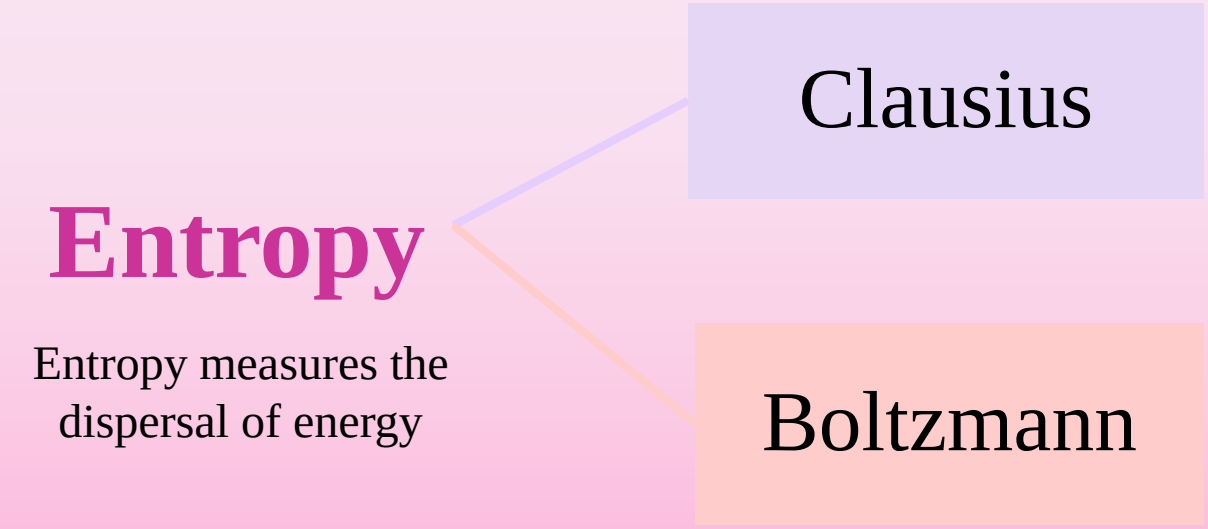
Examples:





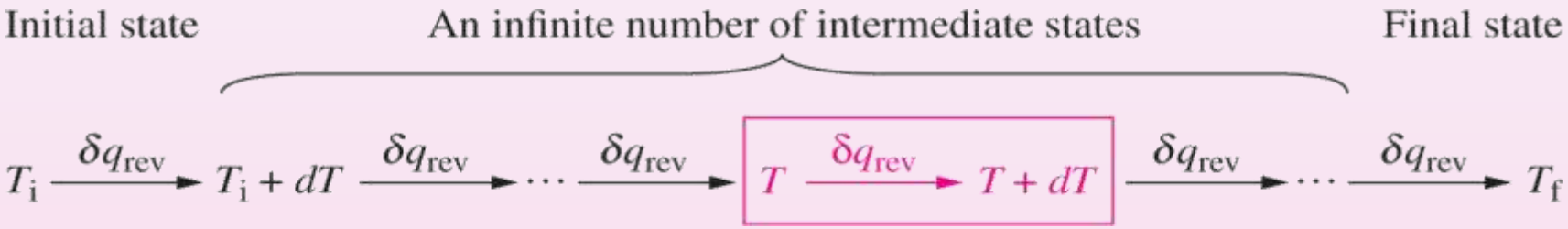
Entropy

- In some chemical and physical processes that proceeded in a certain direction spontaneously, the enthalpy, *H*, of the system increased, decreased or stayed the same. The enthalpy change, ΔH, is not a reliable criterion for deciding whether or not a particular change will occur spontaneously.
- Energy spontaneously "spreads out" or "disperses" unless it is prevented from doing so. ***Entropy*** (S), is a thermodynamic function which measures the dispersal of energy (molecular randomness or disorder).
- The driving force for a spontaneous process is an increase in the entropy of the universe.



Entropy Change: Clausius's View

- Rudolf Clausius discovered ***entropy*** and proved that changes in *S* could be related to ***heat transfer***, provided heat was transferred in a "***reversible***" way.
- For example, to raise the temperature of a system from T_i to T_f in a reversible way means that the temperature is increased by infinitesimal amounts dT, until finally T_f is reached.



- In each step of this reversible process, the entropy changes by a small amount, *dS*, given by:

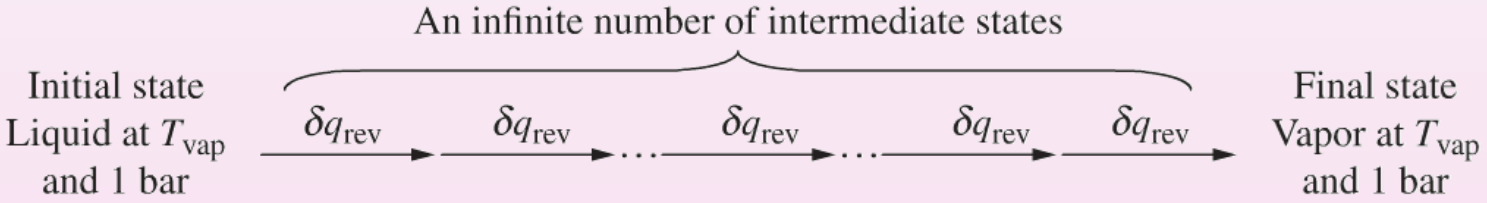
$$dS = \frac{\delta q_{rev}}{T}$$

(where T is the temperature just before delivering the small quantity of heat, δq_{rev} .)

$$\Delta S = \int \frac{\delta q_{rev}}{T} = \frac{1}{T} \int \delta q_{rev} = \frac{q_{rev}}{T}$$

Phase Transitions

- The expression $dS = \delta q_{rev}/T$ can be used to obtain an equation for ΔS when a pure substance undergoes a phase change at constant T.
- Consider the constant pressure vaporization of a liquid at its T_{vap}. The following shows that to carry out the vaporization in a reversible way, heat must be delivered in infinitesimal amounts.



- As The total heat transferred is $q_{vap} = \Delta_{vap}H^{\ominus}$ $\Delta_{vap}S = \dfrac{q_{vap}}{T_{vap}} = \dfrac{\Delta_{vap}H^{\ominus}}{T_{vap}}$
- The equation above can be written more generally as $\Delta_{tr}S = \dfrac{\Delta_{tr}H^{\ominus}}{T_{tr}}$

Phase Transitions

Example: H_2O (s, 1 atm, 273.15K) $\underset{\text{freezing}}{\overset{\text{fusion}}{\rightleftharpoons}}$ H_2O (l, 1 atm, 273.15K)

$$\Delta_{fus}S = \frac{\Delta_{fus}H^{\ominus}}{T_{fus}} = \frac{6.02 \times 10^3 \text{J mol}^{-1}}{273.15 \text{ K}} = 22.0 \text{ J mol}^{-1} \text{ K}^{-1}$$

❖ A useful generalisation known as **Trouton's rule** states that for many liquids at their normal boiling points, **$\Delta_{vap}S°$** has a value of **~87 J mol^{-1}K^{-1}**.

$$\Delta_{vap}S^{\circ} = \frac{\Delta_{vap}H^{\ominus}}{T_{vap}} \approx 87 \text{ Jmol}^{-1}\text{K}^{-1}$$

Examples:

- For benzene (C_6H_6), $\Delta_{vap}S°$ = 87.1 J mol^{-1}K^{-1} ;
- For octane (C_8H_{18}), $\Delta_{vap}S°$ = 86.2 J mol^{-1}K^{-1}.
- Note that the Trouton's rule does **not** apply to liquids with hydrogen bonding.

Entropy Change at Constant Pressure or Volume

❖ ΔS at *constant pressure* without a phase change, from T_i to T_f

$$dS = \frac{\delta q_{rev}}{T}$$

$$dS = (C_p/T)dT$$

$$\Delta S \approx C_p \ln\left(\frac{T_f}{T_i}\right)$$

$$q_p = \Delta H = C_p \Delta T$$

$$\delta q_{rev} = C_p dT$$

❖ ΔS at *constant volume* without a phase change, from T_i to T_f

$$dS = \frac{\delta q_{rev}}{T}$$

$$dS = (C_v/T)dT$$

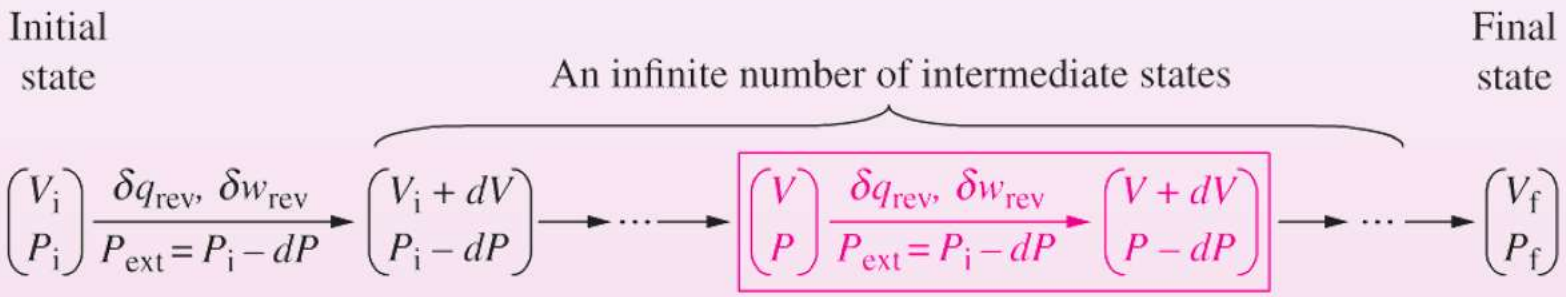
$$\Delta S \approx C_v \ln\left(\frac{T_f}{T_i}\right)$$

$$q_v = \Delta U_v = C_v \Delta T$$

$$\delta q_{rev} = C_v dT$$

Entropy Change at Constant Temperature

- An expression for ΔS for an ideal gas that is expanded or compressed isothermally from V_i to V_f can be obtained. During the expansion or compression, P_i is changed to P_f.
- The reversible expansion of a gas can be represented by the following process.



$$w = -nRT \ln \frac{V_f}{V_i}$$

$$q = nRT \ln \frac{V_f}{V_i}$$

$$dS = \frac{\delta q_{rev}}{T} = \left(\frac{nR}{V}\right) dV$$

$$\Delta S = nR \ln \left(\frac{V_f}{V_i}\right)$$

isothermal volume change for an ideal gas

Since the temperature is constant,

$$P_f V_f = P_i V_i \text{ or } V_f / V_i = P_i / P_f$$

$$\Delta S = nR \ln \left(\frac{V_f}{V_i}\right)$$

$$\Delta S = nR \ln \left(\frac{P_i}{P_f}\right) = -nR \ln \left(\frac{P_f}{P_i}\right)$$

isothermal pressure change for an ideal gas

Summary: Equations for Calculating Enthalpy and Entropy Changes

Process	Enthalpy Change	Entropy Change
Phase transition at the usual (normal) transition temperature	$\Delta_{tr}H^\circ$	$\Delta_{tr}S = \dfrac{\Delta_{tr}H^\circ}{T_{tr}}$
Heating or cooling at constant pressure (no phase transition; C_p has constant value)	$\Delta H = C_p\Delta T$	$\Delta S = C_p \ln\left(\dfrac{T_f}{T_i}\right)$
Heating or cooling at constant volume (no phase transition; C_v has constant value)	$\Delta H = C_v\Delta T + V\Delta P$ $= \Delta U + nR\Delta T$	$\Delta S = C_v \ln(\dfrac{T_f}{T_i})$
Isothermal reversible gas expansion or compression (no phase transition)	$\Delta H = 0$	$\Delta S = nR \ln\left(\dfrac{V_f}{V_i}\right) = -nR \ln\left(\dfrac{P_f}{P_i}\right)$

(NOTE: Each equation for ΔS was obtained by applying $dS = \dfrac{\delta q_{rev}}{T}$)

Example 1

Calculating the Entropy Change for Heating Ice Under Constant Pressure

Calculate the entropy change for the following constant pressure process.

$$H_2O(s, 100\ g, -10\ °C, 1\ bar) \longrightarrow H_2O(l, 100\ g, 10\ °C, 1\ bar)$$

(Given The molar heat capacities of ice and water are, respectively, $C_{p,m}$ = 37.12 J mol^{-1} K^{-1} and $C_{p,m}$ = 75.3 J mol^{-1} K^{-1} . The enthalpy of fusion for ice is $\Delta_{fus}H^{\ominus}$ = 6.01 kJ mol^{-1} at 0 °C.)

Suggested Solution:

The amount of H_2O is $n = (100\ g/18.02\ g\ mol^{-1}) = 5.55$ mol. For the reversible heating of ice from –10 °C to 0 °C:

$$\Delta S_1^\circ = C_p(\text{ice}) \ln\left(\frac{T_f}{T_i}\right) = (5.55\ \text{mol})(37.12\ \text{J mol}^{-1}\ \text{K}^{-1}) \ln\left(\frac{273\ \text{K}}{263\ \text{K}}\right) = 7.69\ \text{J K}^{-1}$$

For the reversible melting of ice at 0 °C:

$$\Delta S_2^\circ = \frac{\Delta_{fus}H^\circ}{T_{fus}} = \frac{(5.55\ \text{mol})(6010\ \text{J mol}^{-1})}{273\ \text{K}} = 122.18\ \text{J K}^{-1}$$

For the reversible heating of water from 0 °C to 10 °C:

$$\Delta S_3^\circ = C_p(\text{water}) \ln\left(\frac{T_f}{T_i}\right) = (5.55\ \text{mol})(75.3\ \text{J mol}^{-1}\ \text{K}^{-1}) \ln\left(\frac{283\ \text{K}}{273\ \text{K}}\right) = 15.03\ \text{J K}^{-1}$$

The total entropy change is the sum of the individual steps:

$$\Delta S_{total}^\circ = \Delta S_1^\circ + \Delta S_2^\circ + \Delta S_3^\circ = (7.69 + 122.18 + 15.03)\ \text{J K}^{-1} = 144.90\ \text{J K}^{-1}$$



Class Practice 1

Calculating the Entropy Change for Isothermal Expansion of an Ideal Gas

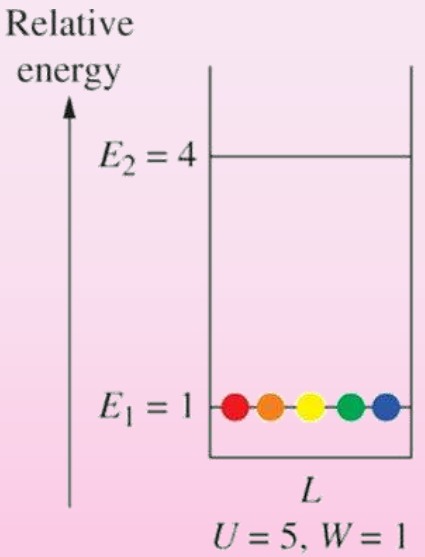
One mole of neon gas, initially at 300 K and 1.00 bar, expands isothermally against a constant external pressure of 0.50 bar until the gas pressure is also 0.50 bar. What is ΔS for the gas?

Entropy Change: Boltzmann's View

Microstates

- Boltzmann's view is a statistical approach using microstates to refine the concept of entropy
- According to quantum mechanics, a **microstate** is a specific microscopic configuration describing how the particles of a system are distributed among the available energy levels.

- Consider a system of five particles confined to a 1-d box of length *L*, we can use the expression, $E_n = \frac{n^2 h^2}{8mL^2}$, to calculate a few energy levels.



- The distribution of five different particles among the particle-in-a-box energy levels. The energies are expressed as multiples of $h^2/8mL^2$.

Entropy Change: Boltzmann's View

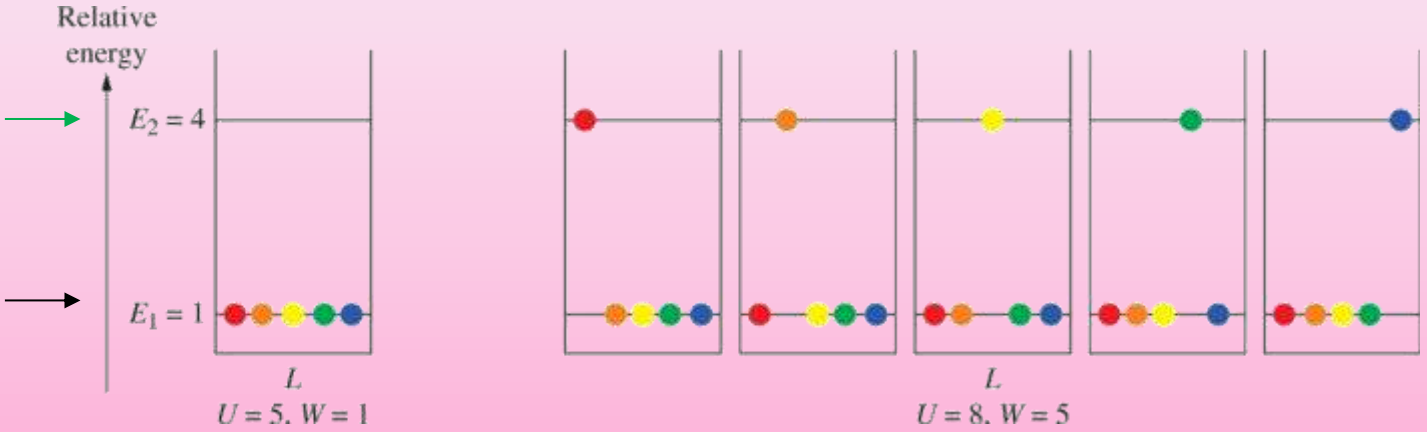
Microstates

- Consider placing the five particles among these energy levels, with the constraint that the total energy, ***U***, of the system must be $5 \times (h^2/8mL^2)$**,** so the only possible arrangement ($W = 1$) has all the particles in the n = 1 level.

$$U = (1 + 1 + 1 + 1 + 1) \times (h^2/8mL^2) = 5 \times (h^2/8mL^2)$$

- Suppose the total energy of the system is increased to $\mathbf{8 \times (h^2/8mL^2)}$ by increasing the temperature, with the length of the box remains the same. There are now five possible microstates ($W = 5$) and an increase in the number of energy levels accessible to the particles.

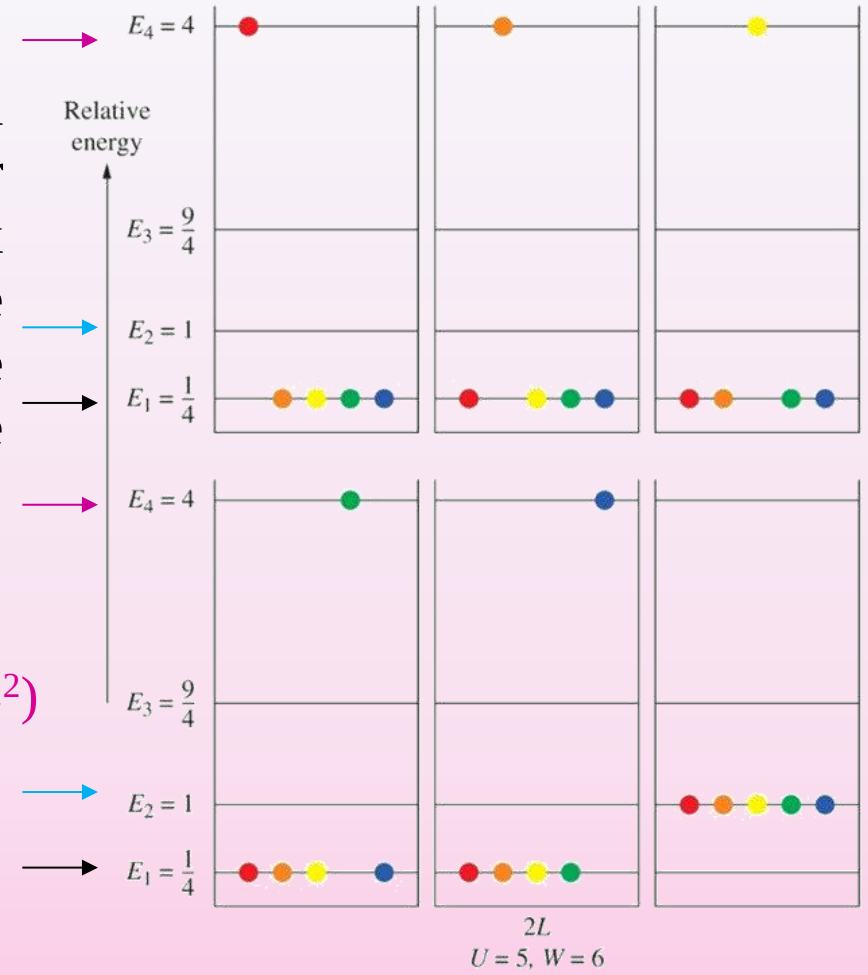
$$U = (1 \times 4 + 4) \times (h^2/8mL^2) = 8 \times (h^2/8mL^2)$$



Entropy Change: Boltzmann's View

Microstates

- When length of the box is increased to **2*L***, the number of microstates for ***U* = 5 ($h^2/8mL^2$)** increases to six (***W* = 6**). More energy levels are accessible because the energy levels are shifted to lower values and are more closely spaced.



5 *W*:

$U = (1/4 \times 4 + 4) \times (h^2/8mL^2) = 5 \times (h^2/8mL^2)$

1 *W*:

$U = (1 \times 5) \times (h^2/8mL^2) = 5 \times (h^2/8mL^2)$

- The **number of microstates (*W*)** and **number of accessible energy levels** increase with both the total energy (multiples of $h^2/8mL^2$) and total space (length of the box) available to the particles of the system.

Boltzmann Equation for Entropy

- Entropy, *S*, is a thermodynamic property that is related to the way in which the energy of a system is distributed among the available energy levels.
- Ludwig Boltzmann associated the number of energy levels in the system with the number of ways of arranging the particles in these energy levels. Boltzmann derived the relationship:

$$S = k_B \ln W$$

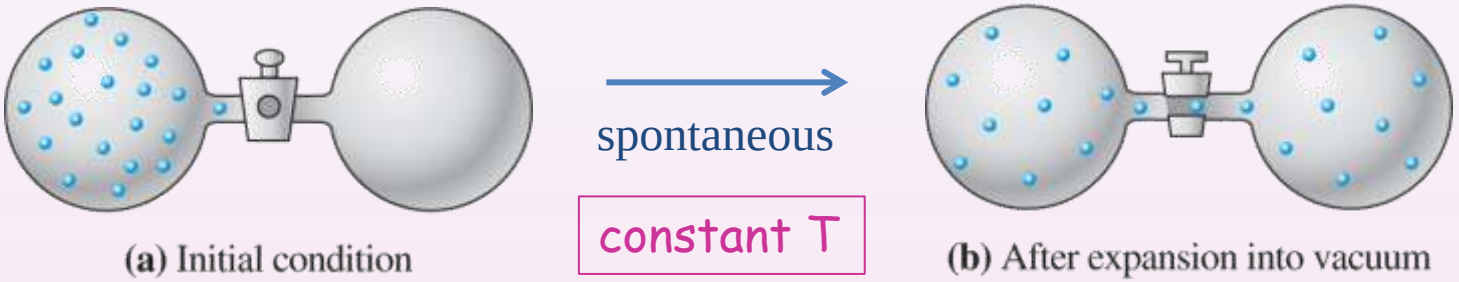
S is a measure of the dispersal of energy:
$\uparrow W \rightarrow \uparrow S$

where *S* is the entropy,
k_B is the Boltzmann constant = R/N_A (1.3807×10^{-23} J K^{-1}),
W is the number of microstates.

- When the *space available to the particles of a system is fixed*, *W* and *S* increase as the total energy, *U*, increases or as *T* increases.
- When the *total energy of a system is fixed*, *W* and *S* increase as the space available to the particles increases.

Microscopic Interpretation of Entropy Change

Figure. 2 Expansion of an ideal gas into a vacuum



(a) Initially, an ideal gas is confined to the bulb on the left at 1.00 bar pressure.

(b) When the stopcock is opened, the gas expands into the identical bulb on the right. **The final condition is one in which the gas is equally distributed between the two bulbs at a pressure of 0.50 bar**.

Why the gas spontaneously expands into a vacuum?

In isothermal expansion, $\Delta U = 0$, $\Delta H = 0$.

Since W is greater for the system in the larger V, ΔS increases

$$\Delta S = k_B \ln W_f - k_B \ln W_i = k_B(\ln W_f - \ln W_i) > 0$$

Microscopic Interpretation of Entropy Change

Figure 3 The spontaneous mixing of ideal gases



The total volume of the system and the total gas pressure remain fixed. The net change is that

(a) before mixing, each gas is confined to half the total volume (a single bulb) at a pressure of 1.00 bar, and

(b) after mixing, each gas has expanded into the total volume (both bulbs) and exerts a partial pressure of 0.50 bar.

$$A(g) + B(g) \longrightarrow \text{mixture of } A(g) \text{ and } B(g)$$

❖ As $\Delta U = 0$ for an isothermal process involving an ideal gas,

$\therefore\ \Delta U_A = \Delta U_B = 0$. Also $\Delta S_A > 0$ and $\Delta S_B > 0$.

$\therefore\ \boldsymbol{\Delta U = \Delta U_A + \Delta U_B = 0}$ and $\boldsymbol{\Delta S = \Delta S_A + \Delta S_B > 0}$

Example 2

Relating Changes in the Number of Microstates to a Volume Change

A system containing four neon atoms is confined to a one-dimensional box. The system undergoes an expansion from 905 pm to 1810 pm at a fixed total energy of 14.0×10^{-24} J.

(a) Determine the number of microstates for both the initial and final states of the system by illustrating, in the manner of Figure 1, the various possibilities for distributing the atoms among the particle-in-a-box energy levels. Use a different color for each atom.

(b) Calculate the entropy change for the system.

Suggested Solution:

(a) Before using the particle-in-a-box energy level expression, we need to express the mass of a neon atom in the SI unit of kilograms.

$$m = 20.18\,\frac{\text{g}}{\text{mol}} \times \frac{10^{-3}\,\text{kg}}{1\,\text{g}} \times \frac{1\,\text{mol}}{6.022 \times 10^{23}\,\text{atoms}} = 3.351 \times 10^{-26}\,\frac{\text{kg}}{\text{atom}}$$

$$E_n = \frac{n^2 h^2}{8mL^2}$$

The energy levels for the 905 pm and 1810 pm boxes are given by

$$L = 905\,\text{pm}: \quad E_n = \frac{(6.626 \times 10^{-34}\,\text{J})^2}{(8)(3.351 \times 10^{-26}\,\text{kg})(905 \times 10^{-12}\,\text{m})^2} \times n^2 = (2.00 \times 10^{-24}\,\text{J}) \times n^2$$

$$L = 1810\,\text{pm}: \quad E_n = \frac{(6.626 \times 10^{-34}\,\text{J})^2}{(8)(3.351 \times 10^{-26}\,\text{kg})(1810 \times 10^{-12}\,\text{m})^2} \times n^2 = (0.500 \times 10^{-24}\,\text{J}) \times n^2$$

Example 2 (Continued)

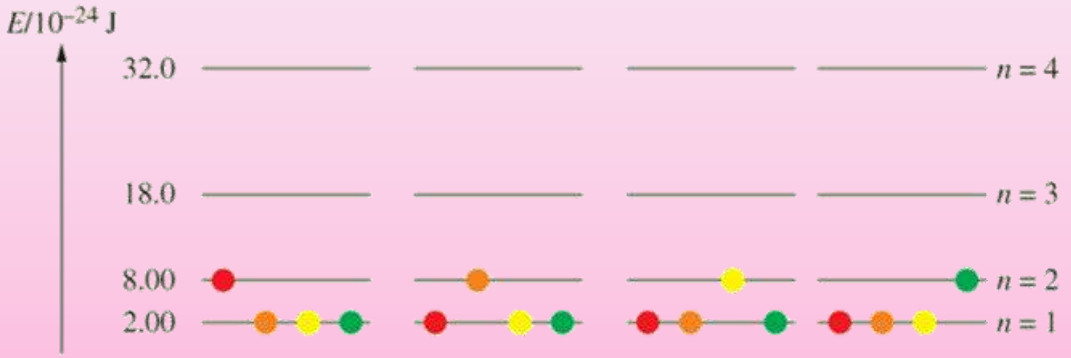
Suggested Solution:

The energies of the first five levels are

	$E_n/10^{-24}$ J	
n	$L = 905$ pm	$L = 1810$ pm
1	2.00	0.500
2	8.00	2.00
3	18.0	4.50
4	32.0	8.00
5	50.0	12.5

As shown in the diagrams that follow, there are 4 microstates for the system in the 905 pm box, and 12 microstates for the system in the 1810 pm box, each having a total energy of 14.0×10^{-24} J.

For the 905 pm box: 4 microstates

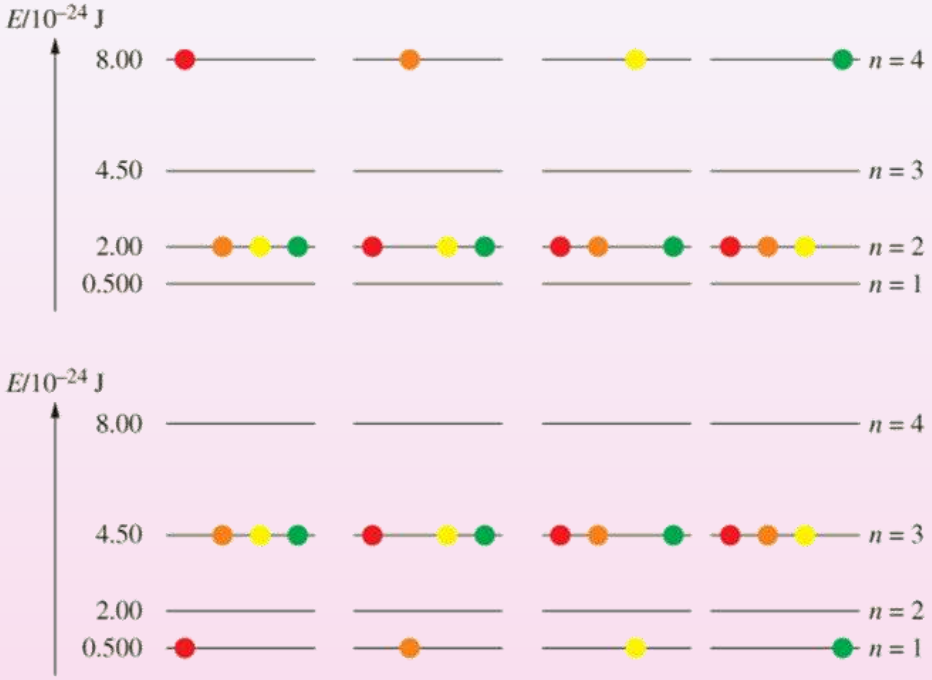


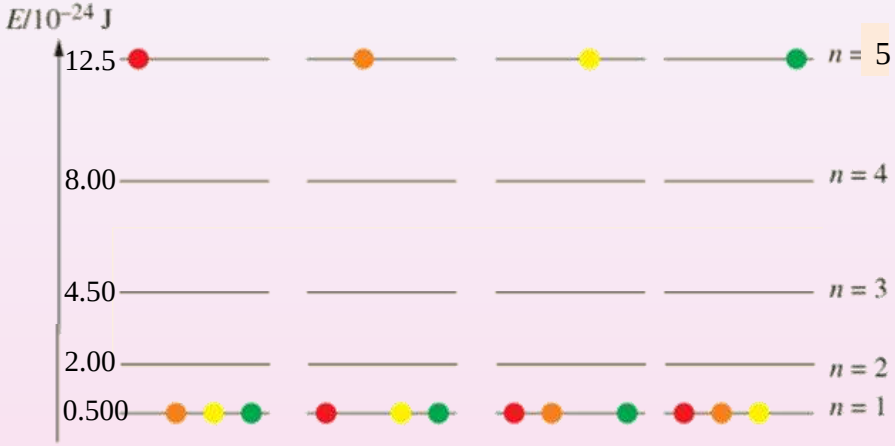
Total energy
= 14.0×10^{-24} J

Example 2 (Continued)

Suggested Solution:

For the 1810 pm box: 12 microstates





(b) The entropy change for the system is

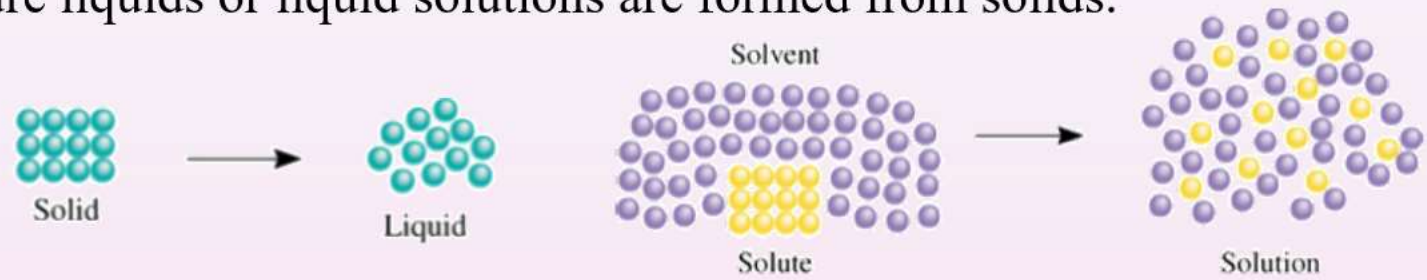
$$\Delta S = k_B \ln 12 - k_B \ln 4 = k_B(\ln 12 - \ln 4) = k_B \ln (12/4) = k_B \ln 3$$

$$= 1.3807 \times 10^{-23}\,\text{J/K} \times 1.099 = 1.52 \times 10^{-23}\,\text{J/K}$$

Increasing Entropy at Different Processes (Revision)

❖ There are four situations generally produce an *increase* in entropy:

i. Pure liquids or liquid solutions are formed from solids.



ii. Gases are formed from either solids or liquids.



iii. The **number of molecules of gas *increases*** as a result of a chemical reaction.

$$2\ K\ (s) + 2\ H_2O\ (l) \rightarrow 2\ KOH\ (aq) + H_2\ (g)$$

$$2\ NaN_3\ (s) \xrightarrow{\Delta} 2\ Na\ (s) + 3\ N_2\ (g)$$

iv. The **temperature of a substance *increases***.



Class Practice 3

Making Qualitative Predictions of Entropy Changes in Physical and Chemical Processes

Predict and briefly explain whether each of the following processes involves an increase or a decrease in entropy or whether the outcome is uncertain.

(a) The decomposition of ammonium nitrate (a fertilizer and a highly explosive compound):

$$2\ NH_4NO_3\ (s) \rightarrow 2\ N_2\ (g) + O_2\ (g) + 4\ H_2O\ (g)$$

(b) The conversion of SO_2 to SO_3 (a key step in the manufacture of sulphuric acid in Contact Process):

$$2\ SO_2\ (g) + O_2\ (g) \overset{V_2O_5,\ 450\ ^oC,\ 1\ atm}{\rightleftharpoons} 2\ SO_3\ (g)$$

(c) The extraction of sucrose from cane sugar juice:

$$C_{12}H_{22}O_{11}\ (aq) \rightarrow C_{12}H_{22}O_{11}(s)$$

(d) The "water gas shift" reaction (involved in the gasification of coal):

$$CO\ (g) + H_2O\ (g) \overset{\text{catalyst}}{\rightleftharpoons} CO_2\ (g) + H_2\ (g)$$



Class Practice 3 (Continued)

Entropy Change (Revision)

- An entropy change is based on two measurable quantities: heat (q) and temperature (T).

$$\Delta S = \frac{q_{rev}}{T}$$

- Unlike U and H, an **absolute** value of the entropy of a substance can be established.
- The absolute entropy of one mole of a substance in its standard state is called the **standard molar entropy ($S^{\ominus}$).**
- The **entropy change of reaction** ($\Delta_r S^{\ominus}$) can be calculated from the standard molar entropies of the substances involved in the reaction as follows:

$$a\,\mathrm{A} + b\,\mathrm{B} + \ldots \longrightarrow c\,\mathrm{C} + d\,\mathrm{D} + \ldots$$

$$\Delta_r S^\circ = \underbrace{[c\,S^\circ_\mathrm{C} + d\,S^\circ_\mathrm{D} + \ldots]}_{\text{Weighted sum of } S^\circ \text{ values for products}} - \underbrace{[a\,S^\circ_\mathrm{A} + b\,S^\circ_\mathrm{B} + \ldots]}_{\text{Weighted sum of } S^\circ \text{ values for reactants}}$$

$$\Delta_r S^{\ominus} = \sum v_p\, S^{\ominus}(\text{products}) - \sum v_r\, S^{\ominus}(\text{reactants})$$

(where v_p is the stoichiometric coefficient of product and v_r is the stoichiometric coefficient of reactant)

Class Practice 4

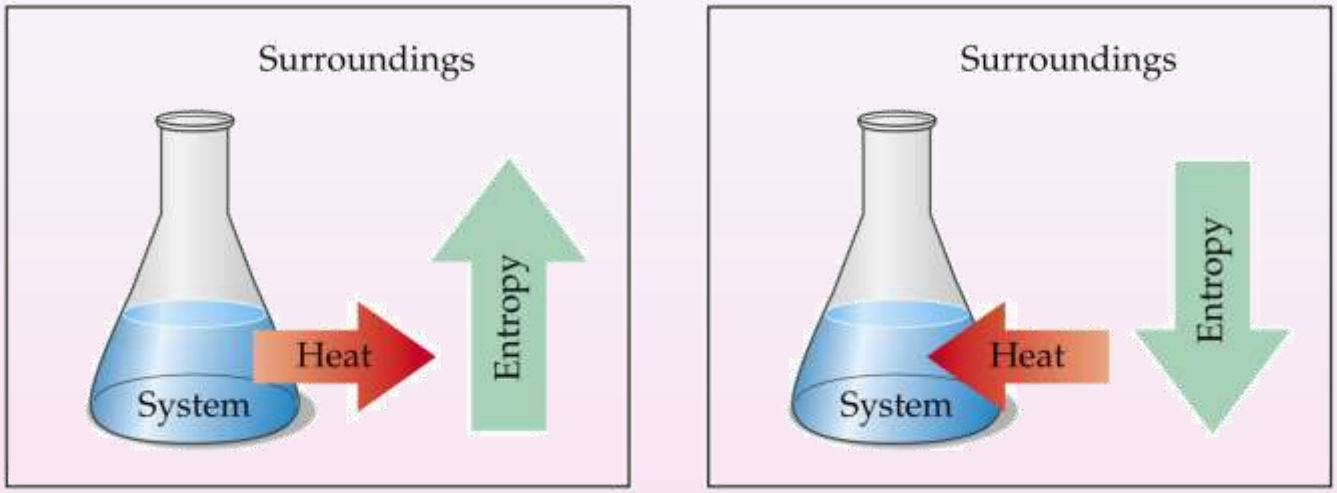
Calculating the Standard Entropy of Reaction

Calculate the standard reaction entropy at 298.15 K for the conversion of nitrogen monoxide to nitrogen dioxide in Ostwald process. (a step in the manufacture of nitric acid).

$$2\,NO(g) + O_2(g) \longrightarrow 2\,NO_2(g) \qquad \Delta_r S^\circ = ?$$

Chemical	NO_2 (g)	NO (g)	O_2 (g)
S^o / J $mol^{-1}K^{-1}$	240.1	210.8	205.1

Criteria for Spontaneity: The Second Law of Thermodynamics (Revision)



Universe = System + Surroundings

The **Second Law of Thermodynamics** states that

$$\Delta S_{universe} = \Delta S_{system} + \Delta S_{surroundings} > 0 \text{ for spontaneity}$$

→ All spontaneous processes produce an **increase** in the entropy of the **universe**.

The Second Law of Thermodynamics and Spontaneity (Revision)

$$\Delta S_{\text{universe}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}}$$

- ΔS_{system} is obtained from the below equation.

$$\Delta_r S^{\ominus} = \sum v_p \, S^{\ominus}(\text{products}) - \sum v_r \, S^{\ominus}(\text{reactants})$$

- $\Delta S_{\text{surroundings}}$ is obtained as shown in the following:

$$\Delta S_{surr} = \frac{q_{surr}}{T_{surr}} = \frac{-q_{sys}}{T_{surr}} = \frac{-\Delta H}{T}$$

$$\boldsymbol{\Delta S_{uni} = \Delta S_{sys} + \Delta S_{surr} = \Delta S_{sys} + \frac{-\Delta H_{sys}}{T}}$$

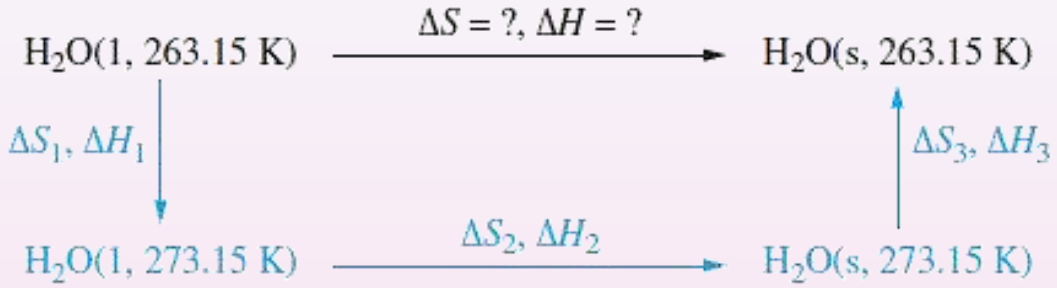
If $\Delta S_{\text{uni}} > 0$, the process is spontaneous;

If $\Delta S_{\text{uni}} < 0$, the process is non-spontaneous;

If $\Delta S_{\text{uni}} = 0$, the process is reversible and the system has reached equilibrium.

Class Practice 5

Determine the enthalpy and entropy change of the freezing of super-cooled water at 263.15 K. Prove that the process is spontaneous.



The molar heat capacities of ice and water are, respectively, $C_{p,m} = 37.12\ J\ mol^{-1}\ K^{-1}$ and $C_{p,m} = 75.3\ J\ mol^{-1}\ K^{-1}$. The enthalpy of fusion for ice is $\Delta_{fus}H^\circ = 6.01\ kJ\ mol^{-1}$ at 0 °C.



Class Practice 5 (Continued)

The Third Law of Thermodynamics (Revision)

- The **Third Law of Thermodynamics** states that the entropy of a *pure, perfectly crystalline substance* is **zero** at the *absolute zero of temperature*.

- At absolute zero, temperature = 0 K, while pure, perfectly crystalline substance has only 1 microstate (W)

$$S = k \ln W$$

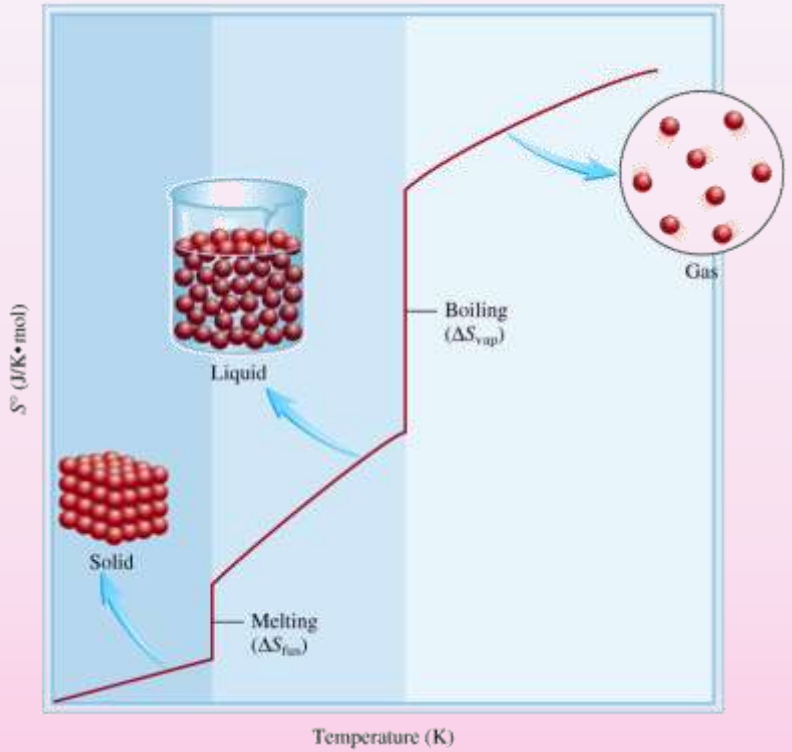
At 0 K, $W = 1$

$$\therefore S = 0$$

$$S(0) = 0 \text{ JK}^{-1}\text{mol}^{-1}$$

$$S(T) = S(0) + \Delta S$$

(from $T_i = 0$ and $T_f = T$)



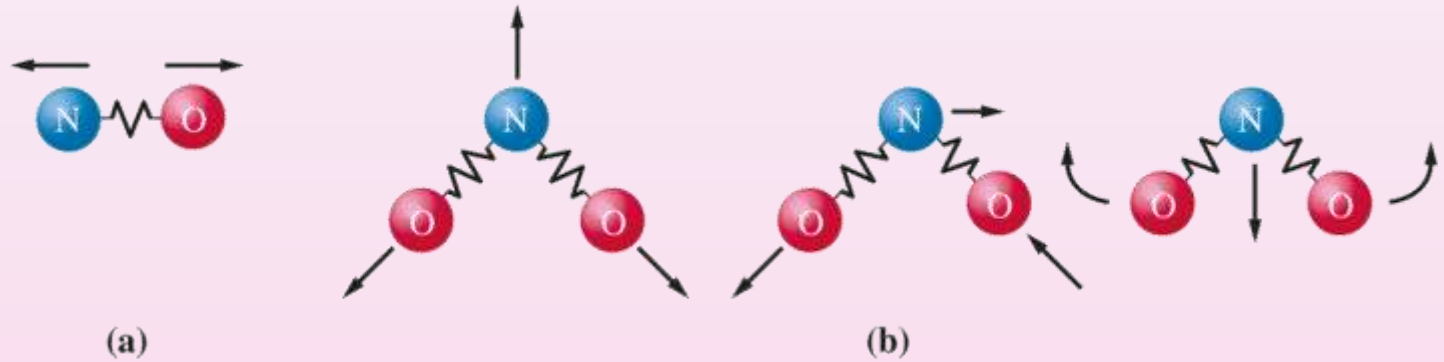
$\Delta S = S_f^{\ominus} - S_i^{\ominus}$ As $S_i^{\ominus} = 0$, $\Delta S = S_f^{\ominus}$

Therefore, for any substance, a specific value to $S°$ at T_f can be assigned.

Combining Boltzmann's and Clausius's Ideas: Absolute Entropies

- The absolute entropy of one mole of substance in its standard state is called the **standard molar entropy, $S^{\ominus}$.**

- Standard molar entropy, $S°$, increases as molecular complexity increases (i.e. as the number of atoms per molecule increases).



Vibrational energy and entropy

The movement of atoms is suggested by the arrows.

(a) The NO molecule has only one type of vibrational motion, whereas

(b) the NO_2 molecule has three.

This difference helps account for the fact that the molar entropy of $NO_2(g)$ is greater than that of NO(g).

Absolute Entropies

Example:

- Consider the following process for a hypothetical substance *X*, which is assumed to be a solid at 0 K and 1 bar and a gas at 298.15 K and 1 bar.

$$\underset{S_i^\circ = 0}{X(s, 0\ K, 1\ bar)} \longrightarrow \underset{S_f^\circ = ?}{X(g, 298.15\ K, 1\ bar)}$$

- This constant pressure process involved the following sequence of changes, where T_{low} represents a very low T (as 0 K is unattainable practically).

1. Heating the solid from T_{low} to its melting point, T_{fus}.

$$X(s, T_{low}, 1\ bar) \longrightarrow X(s, T_{fus}, 1\ bar) \qquad \Delta S_1^\circ \approx C_p(\text{solid}) \ln (T_{fus}/T_{low})$$

2. Melting the solid at T_{fus}.

$$X(s, T_{fus}, 1\ bar) \longrightarrow X(l, T_{fus}, 1\ bar) \qquad \Delta S_2^\circ = \Delta_{fus}H^\circ/T_{fus}$$

3. Heating the liquid from T_{fus} to its boiling point, T_{vap}.

$$X(l, T_{fus}, 1\ bar) \longrightarrow X(l, T_{vap}, 1\ bar) \qquad \Delta S_3^\circ \approx C_p(\text{liquid}) \ln (T_{vap}/T_{fus})$$

4. Vaporizing the liquid at T_{vap}.

$$X(l, T_{vap}, 1\ bar) \longrightarrow X(g, T_{vap}, 1\ bar) \qquad \Delta S_4^\circ = \Delta_{vap}H^\circ/T_{vap}$$

5. Heating the vapor from T_{vap} to 298.15 K.

$$X(g, T_{vap}, 1\ bar) \longrightarrow X(g, 298.15\ K, 1\ bar) \qquad \Delta S_5^\circ \approx C_p(\text{gas}) \ln (298.15\ K/T_{vap})$$

By adding the entropy changes for these steps, we get **the total ΔS.**

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